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United States Patent Application Publication

20250277434

Kind Code

A1

Publication Date

September 04, 2025

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Natural Gas Capture From a Well Stream

Abstract

Hydrocarbon gas capture from a well stream can involve recovering light hydrocarbon gases from liquids and solids via a multiphase separator that is upstream of a sand removal system. Pressure is controlled in the multiphase separator to facilitate hydrocarbon gas capture. The light hydrocarbon gases can be recovered for sale or can be used on-site.

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Family ID: 86160583

Appl. No.: 19/210651

Filed: May 16, 2025

Related U.S. Application Data

parent US continuation 18188747 20230323 PENDING child US 19210651

parent US continuation 17821048 20220819 parent-grant-document US 11639656 child US 18188747

Publication Classification

Int. Cl.: E21B43/34 (20060101); B01D19/00 (20060101); B01D21/24 (20060101)

U.S. Cl.:

CPC E21B43/34 (20130101); B01D19/0042 (20130101); B01D19/0063 (20130101); B01D21/2494 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. Non-Provisional patent application Ser. No. 18/188,747, filed on Mar. 23, 2023, which is a continuation of U.S. Non-Provisional patent application Ser. No. 17/821,048, filed on Aug. 19, 2022, now U.S. Pat. No. 11,639,656, and entitled “Natural Gas Capture from a Well Stream”, the disclosures of which are incorporated herein by reference in their entireties.

FIELD OF THE DISCLOSURE

[0002] The present disclosure generally relates to recovering hydrocarbons from a well stream that includes multiple phases of matter.

BACKGROUND

[0003] After drilling, isolating, or treating a wellbore, there exists a phase of well stream flow called flowback. Flowback can occur after drilling the wellbore and before treatment and production, after treatment and before production, or during production. During flowback that is before production, the well stream can contain formation fluids (water, natural gases, condensates, crude oil), treatment fluids, and solids such as sand, rock fragments, and isolation plug remnants. Sand can be produced from a wellbore when sand is used as proppant to fracture a subterranean formation, and sand can also be native to portions of the subterranean formation and be carried or pushed by fluids (e.g., hydrocarbons, proppant sand) that flow into the wellbore. Well streams produced during flowback that is before production can include sand, water, hydrocarbons (e.g., crude oil, nature gases, condensates, or combinations thereof), drill cuttings (rock fragments), isolation plug remnants, or combinations thereof.

[0004] Various techniques exist for separating the components of flowback well streams. These techniques usually flare or vent the gas that is separated from liquid and solid components of the well stream. Flaring involves the combustion of hydrocarbons in the gas, and creates carbon dioxide. There is an ongoing need to reduce the carbon footprint of well stream flowback separations.

SUMMARY

[0005] Disclosed is a process that can include receiving a well stream from a wellbore; removing a first plurality of solids having a particle size greater than a particle size of sand from the well stream to form a first stream; controlling a flow of the first stream to form a second stream including hydrocarbons, water, and sand; separating the second stream in a multiphase separator into a first portion including light hydrocarbons that are in the gas phase and a second portion including liquid hydrocarbons, water, and sand; flowing the first portion from the multiphase separator in a product gas stream, wherein the product gas stream has a back pressure device located therein; and maintaining, by the back pressure device, a pressure in the multiphase separator in a range of from 15.7 psia to about 500 psia.

[0006] Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] For a more complete understanding of this disclosure, reference is now made to the following description, taken in conjunction with the accompanying drawings, in which:

[0008] FIG. 1 illustrates a side cross-sectional view of a wellbore environment in which the disclosed process and apparatus can be used.

[0009] FIG. 2 illustrates a schematic diagram of embodiments of the disclosed apparatus.

[0010] FIG. 3 illustrates a cut-away schematic diagram of an embodiment of the multiphase separator in the apparatus of FIG. 2.

[0011] FIG. 4 illustrates a plan view of an angled plate.

[0012] FIG. 5 illustrates a cut-away schematic diagram of another embodiment of the multiphase separator of the disclosed apparatus.

[0013] FIG. 6 illustrates a cut-away schematic diagram of the embodiment of the multiphase separator in the apparatus of FIG. 5.

DETAILED DESCRIPTION

[0014] “Flowback” as used herein includes fluids that flow from a wellbore following a treatment, for example, in preparation for a subsequent phase of treatment or in preparation for returning the wellbore to production.

[0015] “Wellbore” as used herein refers to a hole formed in a subterranean formation, for example, through drilling. The wellbore can be conventional (vertically oriented) or unconventional (having horizontally oriented portions). The wellbore can have a depth under the surface of up to about 10,000 ft, for example. Horizontal portions of an unconventional wellbore can extend laterally from a vertical portion and through a subterranean formation for a lateral distance that is up to about 10,000 ft. In aspects, a horizontal portion of the wellbore can have 30 to 90 stages.

[0016] “Condensates” as used herein refers to hydrocarbons produced in a well stream that have 5 to 12 carbon atoms, including linear or branched alkanes (e.g., pentane, hexane, heptane and larger alkanes), cyclic alkanes (e.g., cyclohexane), linear or branched alkenes (e.g., pentene, hexene), aromatic compounds (benzene, toluene, xylenes, or combinations thereof), or combinations thereof.

[0017] “Crude oil” as used herein refers to hydrocarbons having 13 or more carbon atoms.

[0018] The disclosed apparatus and process are described in the context of flowback; however, it is contemplated that the process can be used for any phase of well stream flow (e.g., production) in which the separation of liquid (e.g., non-hydrocarbon liquid such as water or chemical additives injected into a well) from a hydrocarbon gas is desired. The disclosed apparatus and process recover gas for sales or on-site use (avoiding purchase of the gas) that would otherwise be flared or vented. Recovery of this gas reduces the carbon footprint of the well operation because the hydrocarbons in the recovered gas are not converted to carbon dioxide in a flare. Thus aspects of the apparatus and process reduced the carbon footprint in flowback and production operations. Moreover, it has been found that recovering light hydrocarbons in the disclosed multiphase separator, which is upstream of equipment in a sand removal system, enables recovery of 95 vol % or more of methane contained in a well stream (e.g., 95 vol % or more can be recovered for embodiments without compression of product gas, and 80 vol % or more can be recovered for embodiments with compression while fueling the compressors with recovered gas). Further still, the particular configuration and operation of the multiphase separator enables valuable gases to efficiently be separated from the liquid and solids of a well stream, in a manner that the gases are of a purity that can be sold to market or used to power equipment at the wellsite (e.g., power generators for compressors or for ASICs for cryptocurrency (e.g., Bitcoin) mining. For example, a pressure is maintained or controlled at within a setpoint range (e.g., within 5 psi of the setpoint pressure) in the multiphase separator, using back pressure devices to facilitate the light hydrocarbon cut (e.g., C4-hydrocarbons) from the heavier fluids (e.g., C5+ hydrocarbons, water, treatment additive(s), sand, or combinations thereof) in the well stream. Additionally, the configuration of the angled plates in the multiphase separator provide a flowpath in the multiphase separator that facilitates light hydrocarbon separation from the other fluids and that has led to recovery of greater than 95 vol % of the light hydrocarbons during flowback that can be sold or used, for example, for power generators for the equipment or for cryptocurrency (e.g., Bitcoin) mining. Additionally still, the operating pressure of the multiphase separator can be maintained above a sales pipeline pressure such that the pressure difference between the multiphase separator and the sales pipeline can push the product gas to the sales pipeline without the aid of compression.

Additionally still, the operating pressure of the multiphase separator can be maintained above an operating pressure of a sand removal system such that the pressure difference between the multiphase separator the sand removal system can push the slurry from the multiphase separator to the sand removal system. Additionally still, the elevated operating pressure (elevated relative to atmospheric pressure) of the multiphase separator disclosed herein can reduce compression requirements for product gas to pass to a sales pipeline having a pressure less than the operating pressure of the multiphase separator. Additionally still, embodiments having the two-float configuration (two level sensors) on the multiphase separator can remove liquid hydrocarbons from the multiphase separator itself separately from a slurry containing water, treatment additive(s), sand, or combinations thereof, without need for sand removal system to have ability to remove the liquid hydrocarbons from the slurry that is received by the sand removal system from the multiphase separator.

[0019] FIG. 1 illustrates a wellbore environment **100** in which the disclosed process and apparatus **120** can be used. The wellbore environment **100** includes a wellbore **102** formed in a subterranean formation **104**. A well head **106** can cap the wellbore **102** at the surface **103** of the well site. A valve tree **108** (e.g., in embodiments, referred to as a Christmas tree) can be mechanically and fluidly connected to the well head **106** such that fluids can flow through the valve tree **108** and the well head **106** into and out of the wellbore **102**, depending on a direction of flow of fluids.

[0020] Treatment equipment **110** can be placed at the surface **103** of the well site. Treatment equipment **110** can be fluidly connected to the valve tree **108** by conduit(s) **112**. Treatment equipment **110**, by example, can include fracturing equipment. In a fracturing example, the fracturing equipment can inject a high-pressure fracturing fluid (e.g., proppant including sand, ceramic beads, glass beads, or combinations thereof) through the conduit **112**, through the valve tree **108**, through the well head **106**, and into the wellbore **102** so as to fracture portion of the subterranean formation **104**. Fracturing the subterranean formation **104** can produce fractures **114**.

[0021] After use of the treatment equipment **110** (e.g., such as to produce fractures **114**), the wellbore environment **100** can be adjusted for production. For example, the valve tree **108** used during treatment might be changed for a valve tree that is suitable for production. Valve tree **108** illustrated in FIG. 1, and those illustrated herein, are illustrated with the same structure for clarity; however, it is contemplated that different valve trees can include any appropriate valve tree configuration for a particular phase of well operations, e.g., a frac tree for fracturing operations, a stimulation tree for well stimulation via stream and/or chemical injection, or a production tree for production of fluids from the wellbore.

[0022] For flowback phase of operation, fluids **116** can be allowed to flow upward through the wellbore **102** to the surface **103**, through the well head **106**, and to the valve tree **108** (e.g., embodied as a suitable production valve tree). A well stream **118** can be fluidly connected to the valve tree **108** and to the apparatus **120** disclosed herein. The produced fluids **116** can flow through the valve tree **108**, through well stream **118**, and to the apparatus **120** of this disclosure. The well stream **118** can include, without limitation, crude oil, natural gases, condensates, water, proppant (e.g., sand), treatment additives (e.g., chemicals injected into the subterranean formation **104** to alter a chemistry within the formation to enhance hydrocarbon recovery), isolation plug remnants, fragments of casing, rock fragments, or combinations thereof. A pressure of the well stream **118** can be in a range of from about 10,000 psia to 15,000 psia, for example.

[0023] The apparatus **120** of the disclosure can be configured to separate the well stream **118** into one or more of a product gas stream **122**, a residual gas stream **124**, a liquid hydrocarbon stream **125**, a liquid aqueous stream **126**, and a solids stream **128**. Additional detail about the apparatus **120** is discussed in the description below.

[0024] FIG. 2 illustrates a schematic diagram of an embodiment of the disclosed apparatus **200**. The apparatus **200** in FIG. 2 can be used as apparatus **120** in FIG. 1, for example. While the apparatus **200** is described with reference to the equipment illustrated in FIG. 2, it should be

appreciated that the action and functionality performed with the equipment illustrated in FIG. 2 can be performed in one or more aspects of a disclosed process.

[0025] The apparatus **200** can include one or more of a plug catcher **210**, a choke manifold **220**, a multiphase separator **230**, one or more back pressure device **240/244**, a slurry control valve **242**, a sand removal system **250**, a flare **260**, compressor(s) **270**, and a control system **280**. While compressors **270** are illustrated in FIG. 2 to compress a gas recovered from the multiphase separator **230**, alternative aspects of the apparatus **200** and process do not utilize compressors **270**, and as such, any alternative process performed may not include compression of gas recovered from the multiphase separator **230**.

[0026] The apparatus **200** is configured to receive the well stream **118** and to produce a product gas stream **122**, a residual gas stream **124**, a liquid aqueous stream **126**, and a solids stream **128**. The apparatus **200** can, in some aspects, additionally product a liquid hydrocarbon stream **125**.

[0027] The plug catcher **210** is fluidly connected to the well stream **118**. The plug catcher **210** is configured to receive the well stream **118** and to separate heavy particulates from the well stream **118** to produce the first stream **212** containing a wellbore fluid and a solids stream **214** containing the heavy particulates. For example, the plug catcher **210** can be used when zone isolation plugs are milled out to allow fluids **116** to flow upward through the wellbore **102** to the surface **103**. The wellbore fluid **116** flows to the plug catcher **210** via the well stream **118**. Heavy particulates removed by the plug catcher **210** can include the isolation plug remnants, and additionally, can include fragments of casing, rock fragments, or combinations thereof. Other solids, such as sand proppant, can be incidentally removed by the plug catcher **210** along with the heavy particulates, with the understanding that sand is primarily allowed to pass through the screen(s) of the plug catcher **210** and is recovered downstream by the sand removal system **250** in the solids stream **128**. An operating pressure of the plug catcher **210** can be in a range of from about 10,000 psia to 15,000 psia, for example. The plug catcher **210** can be embodied as a single barrel with bypass or as a dual barrel plug catcher, for example. A screen in the plug catcher **210** acts as a filter that prevents heavy particulates from passing through the screen. The plug catcher **210** can include one or more valves connected on the heavy particulate side of the screen, for intermittent or continuous removal of the heavy particulates from the barrel of the plug catcher **210**, into the solids stream **214**. The heavy particulates in solids stream **214** can be discharged into a tank, ditch, or onto the ground next to the plug catcher **210**, for example.

[0028] The first stream **212** can include crude oil, natural gases, condensates, water, proppant (e.g., sand), treatment additives (e.g., chemicals injected into the subterranean formation **104** to alter a chemistry within the formation to enhance hydrocarbon recovery), or combinations thereof. In aspects, the pressure of the first stream **212** can be in a range of from about 10,000 psia to 15,000 psia, for example.

[0029] The choke manifold **220** can be fluidly connected to the first stream **212**. The choke manifold **220** can be configured to receive the first stream **212** and to control a flow of fluids received from the first stream **212** to produce a second stream **222** having a flow suitable for introducing the second stream **222** stream to the multiphase separator **230**. The choke manifold **220** is a manifold assembly that can incorporate chokes, valves, and pressure sensors to provide controlled flow of fluids in second stream **222** that are received by the choke manifold **220** from first stream **212**. The choke manifold **220** can include flanged or integrated gate valves, positive chokes, and adjustable chokes, for example.

[0030] In some aspects, the choke manifold **220** can be configured to reduce the pressure of the fluids received from the first stream **212** to a second pressure in the second stream **222**, such that the pressure of the second stream **222** is lower than a pressure of the first stream **212**. To the extent that pressure is reduced in the choke manifold **220**, the pressure is sufficient such that the pressure differential between the pressure at the end of the second stream **222** that is connected to the choke manifold **220** and the pressure in the product gas stream **122** is sufficient to push the fluids in the

second stream **222** into the multiphase separator **230**, where the pressure in the product gas stream **122** is maintained or controlled using the back pressure device **240**, the back pressure device **244**, the control valve **242**, or combinations thereof (explained in more detail below).

[0031] The second stream **222** can include the same components of the first stream **212**, e.g., crude oil, natural gases, condensates, water, proppant (e.g., sand), treatment additives (e.g., chemicals injected into the subterranean formation **104** to alter a chemistry within the formation to enhance hydrocarbon recovery), or combinations thereof. In aspects, the pressure of the second stream **222** can be at least 15.7, 20, 25, 30, 35, 40, 45, or 50 psia and less than 500, 400, 300, 200, 150, 140, 130, 120, 110, 100, 90, 80, 75, or 60 psia. In aspects, the pressure of the second stream **222** can be in range from any minimum pressure to any maximum pressure described for the second stream **222** herein. In aspects, the pressure of the second stream **222** can be equal or about equal to the operating pressure of the multiphase separator **230**, described herein.

[0032] The second stream **222** is configured to fluidly connect with an inlet **233** of the multiphase separator **230**. In some aspects, a bypass line **226** can be fluidly connected with second stream **222**. In these aspects, valve **223** can be included in second stream **222**, and valve **227** can be included in bypass line **226**. In operation of the multiphase separator **230**, valve **223** can be operated in an open position and valve **227** can be operated in closed position. In scenarios where bypass of the multiphase separator **230** is needed, such as for maintenance of the multiphase separator **230**, valve **223** can be actuated to a closed position and valve **227** can be actuated to an open position such that the fluids in the second stream **222** flow through bypass line **226** and bypass the multiphase separator **230**. The bypass line **226** can have an outlet fluidly connected to an inlet **251** of the sand removal system **250**. In FIG. 2, the bypass line **226** combines with the slurry stream **232** before flowing to the sand removal system **250**. In aspects, valve **223** and valve **227** can be electrically, mechanically, or pneumatically actuated between the closed position and the opening. The valves **223** and **227** can be actuated by personnel manually turning the valves **223** and **227** to respective positions; alternatively, one or both of the valves **223** and **227** can be operably coupled (e.g., via wired or wireless lines) to a control system **280** having software programming for sending signals to equipment that electrically, mechanically, or pneumatically actuates the valves **223** and **227**.

[0033] The multiphase separator **230** is fluidly connected to the second stream **222**. The multiphase separator **230** is configured to receive the second stream **222** and to separate the second stream **222** into the product gas stream **122** and into the slurry stream **232**.

[0034] The multiphase separator **230** can be embodied as one or more vessels (one vessel **231** being illustrated in FIG. 2) having an inlet **233**, a first outlet **234a**, and a second outlet **234b**. The second stream **222** is connected to the inlet **233**, the product gas stream **122** is connected to the first outlet **234a**, and the slurry stream **232** is connected to the second outlet **234b** of the multiphase separator **230**. In aspects, the inlet **233** is located on a side **236** of the vessel **231**, the first outlet **234a** is located on a top **237** of the vessel **231**, and the second outlet **234b** is located on a bottom **238** of the vessel **231**.

[0035] The product gas stream **122** can include light hydrocarbon gases, such as methane, ethane, propane, butane, or combinations thereof. In additional aspects, the product gas stream **122** can include other hydrocarbons that are gases at the operating conditions of the multiphase separator **230**. In aspects, the product gas stream **122** can include light hydrocarbon gases in an amount that is greater than 95, 99, 97, 99, or 99 vol % of the total volume in the product gas stream **122**.

[0036] The slurry stream **232** can include liquids and solid phase particulates. The liquids can include crude oil, condensates, water, treatment additives (e.g., chemicals injected into the subterranean formation **104** to alter a chemistry within the formation to enhance hydrocarbon recovery), or combinations thereof. The solid phase particulates can primarily include sand, as well as small fragments of rock and plugs that pass through the screen(s) in the plug catcher **210**. Within the scope of “slurry” as used herein, it is contemplated that the solids can be dispersed in the liquid, uniformly or nonuniformly; alternatively, such as while in the vessel **231** of the multiphase

separator **230**, the slurry can have most of the solids separate from most of the liquid, such as would occur for sand settling to the bottom **238** of the vessel **231** due to a higher density of the sand than the liquids in the vessel **231**, the liquids being on top of the sand (it is contemplated that liquid can be entrained between particles of sand that are in the bulk of sand at the bottom **238** of the vessel **231**). The scope of slurry contemplates the presence of liquids and solids together regardless whether any certain amount of the solids is dispersed or separate from the liquids.

[0037] The product gas stream **122** can include a back pressure device **240**. Back pressure device **240** can be configured to maintain an operating pressure in the multiphase separator **230** (e.g., a pressure in a range of from 15.7 psia to about 500 psia; alternatively, in a range of from 15.7 psia to about 150 psia; alternatively about 50 psia). The back pressure device **240** can be embodied as a control valve configured to control a flow of gas in the product gas stream **122**. The product gas stream **122** can also include a sensor **241**. The sensor **241** can be a transducer configured to measure pressure of the product gas stream **122**, for example. The back pressure device **240** can be operably coupled to the sensor **241** for actuating the back pressure device **240** so as to control a flow of gas through the product gas stream **122**. Controlling the flow of gas through the product gas stream **122** controls the pressure in the multiphase separator **230**. In some aspects, the back pressure device **240** and sensor **241** can be operably (e.g., via wired or wireless signal) coupled to a control system **280** that has software programming for receiving the signal from the sensor **241**, analyzing the signal, and sending a signal to the back pressure device **240** to control the flow of gas in the product gas stream **122**. In aspects, the sensor **241** can be located to measure pressure in the product gas stream **122** upstream of back pressure device **240** or in the vessel **231**.

[0038] The slurry stream **232** can include a control valve **242**. Control valve **242** can be configured to allow or disallow a flow of slurry out of the bottom **238** of the vessel **231** of the multiphase separator **230**. The control valve **242** can be embodied as a control valve configured to control a flow of slurry in the slurry stream **232**. A level sensor **243** can be coupled to the side **236** of the vessel **231** of the multiphase separator **230**. The sensor **243** can be float sensor configured to extend into an interior of the vessel **231** and to float on or near a surface of the slurry in the vessel **231**. The sensor **243** can send (or not send) a signal to the control valve **242** to allow or disallow flow of slurry out of the vessel **231** in the slurry stream **232**, based on the level of slurry in the vessel **231** relative to a setpoint level. For example, the sensor **243** can send a signal to the control valve **242** to actuate to an open position when the float level is above the setpoint level and send another signal to control valve **242** to actuate to a closed position when the float level falls below the setpoint level. The control valve **242** can be operably coupled to the sensor **243** for actuating the control valve **242** so as to allow or disallow a flow of slurry through the slurry stream **232**. In some aspects, the control valve **242** and sensor **243** can be operably (e.g., via wired or wireless signal) coupled to the control system **280** that has software programming for receiving the signal from the sensor **243**, analyzing the signal, and sending a signal to the control valve **242** to allow or disallow flow of slurry in the slurry stream **232**.

[0039] In some aspects, flowing slurry out of the vessel **231** via control valve **242** can temporarily reduce the pressure in the vessel **231** while the control valve **242** is in the open position, such as when the vol % of liquids and solids recovered from the second stream **222** is greater than the vol % of product gas recovered from the second stream **222**. The change in pressure in the vessel **231** can be measured in the vessel **231**, in the product gas stream **122**, in the product gas bypass line **123**, or combinations thereof. In aspects, the sensor **241** can indicate a change in pressure due to slurry removal from the vessel **231**, and if the pressure falls below a setpoint operating pressure for the multiphase separator **230**, the back pressure device **240** can be actuated to a closed position until the pressure in the vessel **231** rises above the setpoint operating pressure.

[0040] The product gas stream **122** can flow a sales pipeline, a vent, a flare, a gas processing unit (e.g., further treatment or processing), a storage tank, or a combination thereof.

[0041] In FIG. 2, the product sales gas stream **122** flows to a gas processing unit for further

treatment via compression. The product sales gas stream **122** is fluidly connected to one or more compressors **270** that are embodied as the gas processing unit. A gas processing unit can additionally or alternatively include a scrubber to remove sulfur-containing compounds from the product sales gas stream **122**. The compressor(s) **270** are configured to receive the product gas stream **122** and to compress the gases to form a compressed gas stream **272**. The compressed gas stream **272** can have a pressure suitable for storage in a storage tank, flow to a sales pipeline, or flow to further treatment, for example.

[0042] In some aspects, a product gas bypass line **123** can be fluidly connected to the product gas stream **122**. The product gas bypass line **123** can be coupled to residual gas stream **124**, to divert gas in the product gas stream **122** to the flare **260**. For example, a stoppage of flow in product gas stream **122** downstream of the back pressure device **240**, a shutdown of equipment that is downstream of the product gas stream **122** (such as compressor(s) **270** in FIG. 2), a stoppage of gas flow in compressed gas stream **272**, or combinations thereof, can cause need to divert product gas flow through product gas bypass line **123**.

[0043] The product gas bypass line **123** can include another back pressure device **244**. In exemplary scenarios where product gas cannot flow downstream of the first back pressure device **240** in the product gas stream **122**, the pressure in the vessel **231** will increase to the setpoint pressure of the back pressure device **244**. That is, the back pressure device **244** can be configured to actuate at a setpoint pressure that is a ± 10 psi difference from a setpoint pressure of the back pressure device **240**. At the point at which pressure increases to the setpoint pressure of the back pressure device **244**, the back pressure device **244** is configured to actuate, or to be actuated by the control system **280**, from the closed position to the open position, to allow flow of product gas through the product gas bypass line **123**. Product gas can flow through the product gas bypass line **123** until the pressure in the product gas bypass line **123** falls below the setpoint for the back pressure device **244**, after which the back pressure device **244** is actuated from the open position to the closed position. Back pressure device **244** can remain in the closed position until another event causes the pressure in the multiphase separator **230** to rise such that the pressure detected in the product gas bypass line **123** exceeds the setpoint pressure for the back pressure device **244**. In aspects, the back pressure device **244** can be embodied as a control valve configured to control a flow of gas in the product gas bypass line **123**. The product gas bypass line **123** can also include a sensor **245** located in the product gas bypass line **123** upstream of the back pressure device **244**. The sensor **245** can be a transducer configured to measure pressure of the product gas bypass line **123**, for example. The back pressure device **244** can be operably coupled to the sensor **245** for actuating the back pressure device **244** so as to control a flow of gas through the product gas bypass line **123**. Controlling the flow of gas through the product gas bypass line **123** can control the pressure in the multiphase separator **230** at the setpoint of the back pressure device **244**. In some aspects, the back pressure device **244** and sensor **245** can be operably (e.g., via wired or wireless signal) coupled to a control system **280** that has software programming for receiving the signal from the sensor **245**, analyzing the signal, and sending a signal to the back pressure device **244** to control the flow of gas in the product gas bypass line **123**.

[0044] In aspects, the sensor **245** can be the same sensor as sensor **241**, e.g., a single sensor senses the pressure for controlling back pressure device **240** and back pressure device **244**. In aspects having a single sensor **241**, the sensor **241** can be located to measure pressure in the product gas stream **122** upstream of back pressure device **240**, the product gas bypass line **123** upstream of the back pressure device **244**, or the vessel **231**.

[0045] When the product gas bypass line **123** is used for flow of product gas, the back pressure device **240** in the product gas stream **122** can be actuated (e.g., by control system **280**) to a closed position. Actuating the back pressure device **240** to a closed position stops a flow of the product gas in the product gas stream **122**. The back pressure device **244** can be actuated (e.g., by control system **280**) to an open position, to allow flow of product gas in product gas bypass line **123**. The

back pressure device **244** can control the flow of product gas in product gas bypass line **123** so as to control and/or maintain the operating pressure in the multiphase separator **230**.

[0046] When flow of product gas should re-start in the product gas stream **122**, the back pressure device **244** in the product gas bypass line **123** can be actuated (e.g., by control system **280**) to a closed position. Actuating the back pressure device **244** to a closed position stops a flow of the product gas in the product gas bypass line **123**. The back pressure device **240** in the product gas stream **122** can be actuated (e.g., by control system **280**) to an open position, to allow flow of product gas in product gas stream **122**. The back pressure device **240** can control the flow of product gas in product gas stream **122** so as to control and/or maintain the operating pressure in the multiphase separator **230**.

[0047] Aspects of the apparatus **200** and process disclosed herein contemplate that flow of product gas can be switched from the product gas stream **122** to the product gas bypass line **123** by actuating back pressure devices **240** and **244** as described above, while maintaining the operating pressure of the multiphase separator **230**.

[0048] Aspects of the apparatus **200** and process disclosed herein additionally or alternatively contemplate that flow of product gas can be switched from the product gas bypass line **123** to the product gas stream **122** by actuating back pressure devices **240** and **244** as described above, while maintaining the operating pressure of the multiphase separator **230**.

[0049] Aspects of the apparatus **200** and process disclosed herein additionally or alternatively contemplate that product gas can flow in both the product gas stream **122** and in the product gas bypass line **123**, such as in scenarios where sudden the flow of product gas in product gas stream **122** need to decrease but not stop (e.g., pipeline capacity temporarily decreases). For example, the back pressure device **244** in the product gas bypass line **123** can be actuated (e.g., by control system **280**) to an open position while keeping the back pressure device **240** in the open position. Actuating the back pressure device **244** to an open position allows the flow of the product gas in the product gas bypass line **123**. The back pressure devices **240** and **244** can jointly control the flow of product gas in product gas stream **122** and the product gas bypass line **123** so as to control and/or maintain the operating pressure in the multiphase separator **230**.

[0050] In some aspects, flowing slurry out of the vessel **231** via control valve **242** can temporarily reduce the pressure in the vessel **231** due to slurry removal from the vessel **231**, such as when the vol % of liquids and solids recovered from the second stream **222** is greater than the vol % of product gas recovered from the second stream **222**. The change in pressure in the vessel **231** can be measured in the vessel **231**, in the product gas stream **122**, in the product gas bypass line **123**, or combinations thereof. In aspects, one or both of sensors **241** and **245** can indicate a change in pressure, and if the pressure falls below a setpoint operating pressure for the multiphase separator **230**, the back pressure device(s) **240** and/or **244** that are/is in open position (e.g., depending whether product gas flows to compressor(s) **270** or to flare **260**, or to both) can be actuated to a closed position until the pressure in the vessel **231** rises above the setpoint operating pressure, at which time the controls can actuate the back pressure device(s) **240** and/or **244** that are/is in closed position back to the open position.

[0051] The slurry removed from the multiphase separator **230** in slurry stream **232** can flow to the sand removal system **250**. An inlet **251** of the sand removal system **250** can be fluidly connected to the slurry stream **232**.

[0052] In some aspects, the sand removal system **250** in FIG. 2 can be configured to receive the slurry in the slurry stream **232** and to separate the slurry into a residual gas stream **124** containing residual gaseous components that were not removed from the slurry in the multiphase separator **230**, a liquid aqueous stream **126**, and a solids stream **128**. In aspects, the residual gas stream **124** contains less than 5, 4, 3, 2, or 1 vol % of the total gases recovered from the second stream **222**. In some aspects, the liquid aqueous stream **126** can contain crude oil, condensates, water, treatment additives (e.g., chemicals injected into the subterranean formation **104** to alter a chemistry within

the formation to enhance hydrocarbon recovery), or combinations thereof. In alternative aspects, the sand removal system **250** can be configured to separate liquid hydrocarbons (e.g., crude oil, condensates, or both) and water received from the slurry stream **232** so as to produce liquid hydrocarbon stream **125** containing the liquid hydrocarbons and the aqueous stream **126** containing water. The solids stream **128** can contain sand, and in some aspects, small fragments of rock and plugs that pass through the screen(s) in the plug catcher **210**.

[0053] The sand removal system **250** can be embodied as any equipment known in the art with the aid of this disclosure that is configured to separate a slurry into components parts. For example, the sand removal system **250** can be embodied to include a gravity separation tank, where water of the slurry can be allowed to evaporate from the solids (e.g., sand). The sand removal system **250** can additionally or alternatively include other techniques for separating components of the slurry received from the multiphase separator **230**, such as techniques disclosed in U.S. Pat. No. 8,002,050 or U.S. Patent Application Publication No. 2013/0284026A1, each of which is incorporated by reference in its entirety.

[0054] The residual gas stream **124** can flow to the flare **260**. The flare **260** can be any flare known in the art with the aid of this disclosure. The flare **260** can have a flare stack and a flare tip for combustion of flammable components (e.g., the residual gas from the residual gas stream **124**, and in some aspects, at least a portion of the hydrocarbons from product gas stream **122**).

[0055] The liquid aqueous stream **126** can flow to a liquid storage tank or to downstream separation stages for further separation of water from other liquids (e.g., liquid hydrocarbons, treatment additives, or both). For example, the sand removal system **250** can include a gravity separation tank. In some aspects, the liquid aqueous stream **126** containing liquid hydrocarbons and water can be recovered from the gravity separation tank and sent to a liquid hydrocarbon/water separation apparatus such as that disclosed in U.S. Patent Application Publication No. 2013/0284026A1, for skimming of the liquid hydrocarbons from water. In other aspects, the sand removal system **250** can include a liquid hydrocarbon/water separation apparatus such as that disclosed in U.S. Patent Application Publication No. 2013/0284026A1, and the sand removal system **250** can produce the liquid aqueous stream **126** containing water and the liquid hydrocarbon stream **125** containing liquid hydrocarbons. The water can be sent to storage or reinjected into the subterranean formation, for example. The liquid hydrocarbons can be recovered for storage or pumped to a sales pipeline.

[0056] Control system **280** can be embodied as one or more computers having hardware and software for receiving signals from sensors and for sending signals or otherwise controlling instrumentation in the apparatus **200** and process. For example, the control system **280** can be programmed to control one or more of valves **223**, **227**, and **242** and back pressure devices **240** and **244**.

[0057] FIG. **3** illustrates a cut-away schematic diagram of an embodiment of the multiphase separator **230** of the disclosed apparatus **200**. Parts of the multiphase separator **230** in FIG. **3** are labeled with the same references numerals for like parts illustrated in FIG. **2**. Angled plates **310**, **320**, and **330** can be seen in the interior of the vessel **231** of the multiphase separator **230**.

[0058] The multiphase separator **230** in FIG. **3** is described with respect to separation of the second stream **222** into a first portion that is in gas phase and a second portion that is in liquid and solid phases. As previously described for the multiphase separator **230** in FIG. **2**, the multiphase separator **230** in FIG. **3** can include a vessel **231**, an inlet **233** connected to a side **236a** of the vessel **231**, a first outlet **234a** connected to a top **237** of the vessel **231**, and a second outlet **234b** connected to a bottom **238** of the vessel **231**. The inlet **233** is configured to receive the second stream **222**, the first outlet **234a** is configured to receive the first portion from the interior of the vessel **231** (gases separated from liquids and solids in the vessel **231**), and the second outlet **234b** is configured to receive the second portion from the interior of the vessel **231** (the liquid and solids separated from the gases in the vessel **231**).

[0059] A first angled plate **310** is connected to an inner wall **239** of the vessel **231** at a first location that is below where the inlet **233** is connected to the side **236a** of the vessel **231**. Particularly, end **312** of the first angled plate **310** is connected to the side **236a** of the vessel **231** below the inlet **233**. The first angled plate **310** extends at an angle downwardly past the longitudinal axis L of the vessel **231**. Particularly opposite end **314** of the first angled plate **310** extends from the inner wall **239** and past the longitudinal axis L. An angle $\theta_{\text{sub.1}}$ between a bottom surface **311** of the first angled plate **310** and the inner wall **239** (e.g., the vertical wall) of the vessel **231** is in a range of from about 45° to about 75°; alternatively, about 60°.

[0060] A second angled plate **320** is connected to the inner wall **239** of the vessel **231** at a second location that is lower than the first location where the first angled plate **310** is connected to the inner wall **239**. Particularly, end **322** of the second angled plate **320** is connected to the side **236b** of the vessel **231** that is opposite the side **236a** where the second angled plate **320** is connected. The second angled plate **320** extends at an angle downwardly past the longitudinal axis L of the vessel **231**. Particularly opposite end **324** of the second angled plate **320** extends from the inner wall **239** and past the longitudinal axis L. An angle $\theta_{\text{sub.2}}$ between a bottom surface **321** of the second angled plate **320** and the inner wall **239** (e.g., the vertical wall) of the vessel **231** is in a range of from about 45° to about 75°; alternatively, about 60°.

[0061] A third angled plate **330** is connected to the inner wall **239** of the vessel **231** at a third location that is lower than the first location where the first angled plate **310** is connected to the inner wall **239** and lower than the second location where the second angled plate **320** is connected to the inner wall **239**. Particularly, end **332** of the third angled plate **330** is connected to the side **236a** of the vessel **231** that is the same side **236a** where the first angled plate **310** is connected and opposite the side **236b** where the second angled plate **320** is connected to the inner wall **239**. The third angled plate **330** extends at an angle downwardly from the inner wall **239** and past the longitudinal axis L of the vessel **231**. Particularly opposite end **334** of the third angled plate **330** extends past the longitudinal axis L. An angle $\theta_{\text{sub.3}}$ between a bottom surface **331** of the third angled plate **330** and the inner wall **239** (e.g., the vertical wall) of the vessel **231** is in a range of from about 45° to about 75°; alternatively, about 60°.

[0062] Alternative aspects of the multiphase separator **230** contemplate that more or fewer angled plates can be used in the vessel **231**.

[0063] In aspects, a distance D1 between the opposite end **314** of the first angled plate **310** and a top surface **323** of the second angled plate **320** is in a range of from about 6 inches to about 12 inches; alternatively, from about 8 inches to about 11 inches; alternatively, from about 9 inches to about 10 inches. In aspects, a distance D1 between the opposite end **324** of the second angled plate **320** and a top surface **333** of the third angled plate **330** is in a range of from about 6 inches to about 12 inches; alternatively, from about 8 inches to about 11 inches; alternatively, from about 9 inches to about 10 inches.

[0064] The level sensor **243** can be seen with float **340** connected to a stem **341** that is connected to the body mount **342**. The body mount **342** is connected to the side **236a** of the vessel **231**. While shown connected to side **236a**, the body mount **342** can be connected to any location on the circumference of the vessel **231**. Generally, the level sensor **243** is connected to the vessel **231** at a location lower than the lowest angled plate (e.g., the third angled plate **330** in FIG. 3). The float **340** and stem **341** are sized so as not to contact the bottom surface of the lowest plate (e.g., bottom surface **331** of the third angled plate **330**). Double arrow A indicates the movement of the float **340** of the level sensor **243**. The level sensor **243** is configured such that the float **340** floats on a top surface of the second portion of fluids in the vessel **231** (including liquid hydrocarbons, treatment additive(s), water, sand, or combinations thereof). For example, for a second portion that contains both solids (e.g., sand) and liquids (e.g., water, liquid hydrocarbons (e.g., crude oil, condensates, or both), treatment additives, or combinations thereof), the sand can fall to the bottom **238** of the vessel **231** and the liquids can locate on top of the sand. The float **340** is configured with a density

less than the liquid hydrocarbons (e.g., condensates, crude oil, or both) such that the float **340** can float on a surface of crude oil, condensates, water, or a mixture thereof that is inside the vessel **231**. The float **340** moves up and down as the level of solids and liquids raises and lowers due to incoming fluid from the second stream **222** and due to removal of the second portion from the vessel **231** via the slurry stream **232**.

[0065] The multiphase separator **230** can be manufactured of suitable metal, flanges, piping, valving, and combinations thereof. Exemplary internal diameter of the vessel **231** of the multiphase separator **230** includes diameters in the range of from about 36 inches to about 60 inches; alternatively, from about 42 inches to about 54 inches; alternatively, about 48 inches. Exemplary heights of the vessel **231** of the multiphase separator **230** includes heights in the range of from about 108 inches to about 132 inches; alternatively, from about 114 inches to about 126 inches; alternatively, about 120 inches. An exemplary location for the level sensor **243** can be a height on the side **236** of the vessel **231** that is about 3 feet above the bottom **238** of the vessel **231**. An exemplary location for the inlet **233** can be a height on the side **236** of the vessel **231** that is about 8 feet above the bottom **238** of the vessel **231**, and the angled plates **310/320/330** are placed at locations in the vessel **231** that have a height above the bottom **238** of the vessel **231** that is less than the height of the inlet **233**.

[0066] Inlet **233** and outlet **234a** can each have a nominal pipe size in a range of from about 4 inches to about 8 inches; alternatively, about 6 inches. Outlet **234b** can have a nominal pipe size that is greater than the inlet **233** and outlet **234a**, for example, in a range of from about 8 inches to about 12 inches; alternatively, about 10 inches. The diameter of outlet **234a** is configured to prevent undue friction pressure losses of product gas stream **122** and also to reduce internal flow velocities of the product gas stream **122** where there could be a potential of entraining a small percentage of the liquid phase in the fluid flow into outlet **234a**. The larger diameter (nominal pipe size) of the outlet **234b** is configured to prevent solids blockage in the outlet **234b**.

[0067] In aspects, the operating conditions in the multiphase separator **230** can include temperature, pressure, residence time of slurry, and throughput. In aspects, no heat is added or removed from the multiphase separator **230**, and the temperature can be the ambient temperature of the fluids as affected by environmental conditions at the wellsite, e.g., 100° F. day time temperature during a summer in eastern Texas. The operating pressure can be any pressure disclosed herein for the multiphase separator **230**, such as at least 15.7, 20, 25, 30, 35, 40, 45, or 50 psia and less than 500, 400, 300, 250, 200, 150, 140, 130, 120, 110, 100, 90, 80, 75, or 60 psia. In aspects, the pressure of the multiphase separator **230** can be in range from any minimum pressure to any maximum pressure described for the multiphase separator **230**. The residence time of the slurry can be the time between actuation of the control valve **242** in the slurry stream **232**, which be in a range of from about 1 minute to 1 day. The throughput of the multiphase separator **230** can depend on the flow rate of the well stream **118**, and the vessel **231**, streams **122** and **232**, back pressure devices **240** and **244**, and control valve **242** can be appropriately sized for the flowrate of fluids in the well stream **118**. An example throughput can be in a range of less than about 5,500 barrels per day.

[0068] Fluid from stream **222** can flow into the vessel **231** via the inlet **233**. Components of the stream **222** that are in gas phase (e.g., light hydrocarbons) separate from the liquids (e.g., condensate, crude oil, water, treatment additives, or combinations thereof). The operating conditions (temperature and pressure) inside the vessel **231** are such that light hydrocarbons, to the extent not already in a gas phase in stream **222**, can flash from the liquids to form the first portion in gas phase that is described herein, recovered in the product gas stream **122**. The liquids and solids together move on the top surface **313** of the first angled plate **310** from end **312** to end **314**, falling down to the top surface **323** of the second angled plate **320** at end **314** of the first angled plate **310**. The liquids and solids together move on the top surface **323** of the second angled plate **320** from end **322** to end **324**, falling down to the top surface **333** of the third angled plate **330** at end **324** of the second angled plate **320**. The end **334** of the third angled plate **330** extends past the

longitudinal axis L of the vessel **231** such that solids and liquids fall to the bottom **238** of the vessel **231** but not directly over the second outlet **234b**.

[0069] Configuring the ends **314/324/334** of each of the plates **310/320/330** to extend past the longitudinal axis L of the vessel **231** creates a single flow path for the fluids that enter the vessel **231**. The fluids must flow over the first angled plate **310** before flowing over the second angled plate **320**, and must flow over the second angled plate **320** before flowing over the third angled plate **330**. The configuration of the plates **310/320/330** increases the length of the flow path compared to vessels where the fluids fall directly to the bottom or that have only one plate which the fluid contacts before falling to the bottom of the vessel. The increased length of the flow path increases the amount of time given for gases to separate from the liquids and solids in the vessel **231** before the liquids and solids fall to the bottom **238** of the vessel **231**. This increased length of flow path has led to recovery of more than insignificant amounts of light hydrocarbons during flowback that can be sold or used, for example, for power generators for the equipment or for cryptocurrency (e.g., bitcoin) mining.

[0070] FIG. **4** illustrates a plan view of an angled plate **310/320/330**. The plate **310/320/330** will be described with additional reference to the reference numerals in and described for FIG. **3**.

[0071] End **312** of the first angled plate **310** is connected (e.g., welded) to the side **236a** of the vessel **231** of the multiphase separator **230**, and opposite end **314** of the plate **310** can be seen extending past the longitudinal axis L of the vessel **231**. End **322** of the second angled plate **320** is connected (e.g., welded) to the side **236b** of the vessel **231** of the multiphase separator **230**, and opposite end **324** of the plate **320** can be seen extending past the longitudinal axis L of the vessel **231**. End **332** of the third angled plate **330** is connected (e.g., welded) to the side **236a** of the vessel **231** of the multiphase separator **230**, and opposite end **334** of the plate **310** can be seen extending past the longitudinal axis L of the vessel **231**.

[0072] Each plate **310/320/330** can have a first portion **401** and a second portion **402** connected to the first portion **401**. First portion **401** and second portion **402** are generally in the same plane. In aspects, the first portion **401** and the second portion **402** are integrally formed; alternatively, the first portion **401** can be formed separately from the second portion **402**, and the side **403** of the first portion **401** can be connected to the side **404** of the second portion **402**.

[0073] The first portion **401** is generally semi-elliptical in shape in the view shown in FIG. **3**. The major radius R1 of the semi-ellipse of the first portion **401** can be in a range of from about 20 inches to about 30 inches; alternatively, from about 24 inches to about 28 inches; alternatively, from about 26 inches to about 27 inches. The minor radius R2 of the semi-ellipse of the first portion **401** can be in a range of from about 18 inches to about 28 inches; alternatively, from about 21 inches to about 26 inches; alternatively, from about 23 inches to about 24 inches. Edge **405** of the first portion **401** can abut and be connected to the inner wall **239** of the vessel **231**.

[0074] The second portion **402** has edges or sides **406** that can abut or be connected to the inner wall **239** of the vessel **231**. End or side **407** of the second portion **402**, that is opposite the side **404** of the second portion **402** is the point where fluids fall downward in the vessel **231** (e.g., to another angled plate below the end or side **407** or to the bottom **238** of the vessel **231**). End or side **407** of the second portion **402** can have a length or distance D2 in a range of from about 18 inches to about 28 inches; alternatively, from about 21 inches to about 24 inches; alternatively, about 22 inches. The second portion **402** can have a width or distance D3 in a range of from about 7 inches to about 10 inches; alternatively, from about 8 inches to about 9 inches.

[0075] Regarding vertically positioning of the angled plates **310/320/330**, the end **314/324/334** is generally position lower in the vessel **231** than end **312/322/332**. Fluids travel or move on the top surface **313/323/333** in the direction of arrow B.

[0076] FIG. **5** illustrates a schematic diagram of another embodiment of the disclosed apparatus **500**. The apparatus **500** in FIG. **5** can be used as apparatus **120** in FIG. **1**, for example. While the apparatus **500** is described with reference to the equipment illustrated in FIG. **5**, it should be

appreciated that the action and functionality performed with the equipment illustrated in FIG. 5 can be performed in one or more embodiments of a disclosed process.

[0077] The apparatus **500** includes the same equipment (e.g., one or more of a plug catcher **210**, a choke manifold **220**, a multiphase separator **230**, one or more back pressure device **240/244**, a slurry control valve **242**, a sand removal system **250**, a flare **260**, compressor(s) **270**, and a control system **280**) as the apparatus **200** in FIG. 2. The apparatus **500** in FIG. 5 differs from the apparatus **200** in FIG. 2 in that the liquid hydrocarbon stream **125** is fluidly connected to the side **236** of the multiphase separator **230** instead of to the sand removal system **250**. The multiphase separator **230** in apparatus **500** is configured to recover liquid hydrocarbons (e.g., condensates, crude oil, or both) in the liquid hydrocarbon stream **125** instead of in the slurry stream **232** as described for apparatus **200** in FIG. 2. Fluids received from second stream **222** into the vessel **231** of the multiphase separator **230** separate into the first portion containing light hydrocarbons in the gas phase and a second portion containing liquid hydrocarbons, water, and sand. The second portion further separate in the vessel **231** into a layer of the liquid hydrocarbons on top of the water, and a layer of water on top of the sand. The liquid hydrocarbons includes condensates, crude oil, or both. The liquid hydrocarbons can float on top of the water because the density of the liquid hydrocarbons is less than a density of water. Any alternative embodiments described for apparatus **200** in FIG. 2 apply to the apparatus **500**, unless such embodiments would be inconsistent with the operation of apparatus **500**.

[0078] The apparatus **500** is configured to receive the well stream **118** and to produce a product gas stream **122**, a residual gas stream **124**, a liquid hydrocarbon stream **125**, a liquid aqueous stream **126**, and a solids stream **128**.

[0079] The plug catcher **210**, choke manifold **220**, multiphase separator **230**, sand removal system **250**, flare **260**, and optional compressor(s) **270** can be connected as described for the apparatus **200** in FIG. 2 and can have same operation except as described below, and as such, the descriptions for apparatus **200** that apply for apparatus **500** are not reproduced here.

[0080] The multiphase separator **230** is fluidly connected to the second stream **222**. The multiphase separator **230** is configured to receive the second stream **222** and to separate the second stream **222** into the product gas stream **122**, the liquid hydrocarbon stream **125**, and the slurry stream **232**.

[0081] The multiphase separator **230** can be embodied as one or more vessels (one vessel **231** being illustrated in FIG. 5) having an inlet **233**, a first outlet **234a**, a second outlet **234b**, and a third outlet **234c**. The second stream **222** is connected to the inlet **233**, the product gas stream **122** is connected to the first outlet **234a**, the slurry stream **232** is connected to the second outlet **234b**, and the liquid hydrocarbon stream **125** is connected to the third outlet **234c** of the multiphase separator **230**. In aspects, the inlet **233** is located on a side **236** of the vessel **231**, the first outlet **234a** is located on a top **237** of the vessel **231**, the second outlet **234b** is located on a bottom **238** of the vessel **231**, and the third outlet **234c** is located on the side **236** of the vessel **231**.

[0082] The product gas stream **122** can include light hydrocarbon gases as described for apparatus **200** in FIG. 2.

[0083] The liquid hydrocarbon stream **125** can include liquid hydrocarbons such as condensates, crude oil, or both. The liquid hydrocarbon stream **125** can be fluidly connected to storage tank(s), and the liquid hydrocarbons can be recovered for storage or pumped to a sales pipeline.

[0084] The slurry stream **232** can include a slurry of water, treatment additives, sand, or combinations thereof that is separated from the liquid hydrocarbons in the vessel **231**. Because any condensates and/or crude oil are recovered in liquid hydrocarbon stream **125**, the slurry stream **232** in apparatus **500** does not contain any significant amount of liquid hydrocarbons (e.g., less than 1, 0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2, or 0.1 wt % liquid hydrocarbons based on a total volume of the slurry stream **232**). Within the scope of “slurry” as used herein, it is contemplated that the solids can be dispersed in the liquid of the slurry, uniformly or nonuniformly; alternatively, such as while in the vessel **231** of the multiphase separator **230**, the slurry can have most of the solids separate

from most of the liquid, such as would occur for sand settling to the bottom **238** of the vessel **231** due to a higher density of the sand than the liquids in the vessel **231**, the liquids being on top of the sand (it is contemplated that liquid can be entrained between particles of sand that are in the bulk of sand at the bottom **238** of the vessel **231**). The scope of slurry contemplates the presence of liquids and solids together regardless whether any certain amount of the solids is dispersed or separate from the liquids.

[0085] The product gas stream **122** can include a back pressure device **240** as described for the apparatus **200**.

[0086] The liquid hydrocarbon stream **125** can include a control valve **246**. Control valve **246** can be configured to allow or disallow a flow of liquid hydrocarbons out of the side **236** of the vessel **231** of the multiphase separator **230**. The control valve **246** can be embodied as a control valve configured to control a flow of liquid hydrocarbons in the liquid hydrocarbon stream **125**. A level sensor **247** can be coupled to the side **236** of the vessel **231** of the multiphase separator **230**. The sensor **247** can be float sensor configured to extend into an interior of the vessel **231** and to float on or near a surface of the liquid hydrocarbons in the vessel **231**. The sensor **247** can send (or not send) a signal to the control valve **246** to allow or disallow flow of liquid hydrocarbons out of the vessel **231** in the liquid hydrocarbon stream **125**, based on the level of liquid hydrocarbons floating on top of the slurry in the vessel **231** relative to a setpoint level for the liquid hydrocarbons. For example, the sensor **247** can send a signal to the control valve **246** to actuate to an open position when the float level is above the setpoint level and send another signal to control valve **246** to actuate to a closed position when the float level falls below the setpoint level. The control valve **246** can be operably coupled to the sensor **247** for actuating the control valve **246** so as to allow or disallow a flow of liquid hydrocarbons through the liquid hydrocarbon stream **125**. In some aspects, the control valve **246** and sensor **247** can be operably (e.g., via wired or wireless signal) coupled to the control system **280** that has software programming for receiving the signal from the sensor **247**, analyzing the signal, and sending a signal to the control valve **246** to allow or disallow flow of liquid hydrocarbons through the liquid hydrocarbon stream **125**.

[0087] The slurry stream **232** can include a control valve **242** as described for apparatus **200** in FIG. 2.

[0088] In some aspects, flowing slurry out of the vessel **231** via control valve **242** and flowing liquid hydrocarbons out of the vessel **231** via control valve **246** can temporarily reduce the pressure in the vessel **231** while the control valve **242** and/or control valve **246** is/are in the open position, such as when the vol % of liquids and solids recovered from the second stream **222** is greater than the vol % of product gas recovered from the second stream **222**. The change in pressure in the vessel **231** can be measured in the vessel **231**, in the product gas stream **122**, in the product gas bypass line **123**, or combinations thereof. In aspects, one or both of sensors **241** and **245** can indicate a change in pressure, and if the pressure falls below a setpoint operating pressure for the multiphase separator **230**, the back pressure device(s) **240** and/or **244** that are/is in open position (e.g., depending whether product gas flows to compressor(s) **270** or to flare **260**, or to both) can be actuated to a closed position until the pressure in the vessel **231** rises above the setpoint operating pressure, at which time the controls can actuate the back pressure device(s) **240** and/or **244** that are/is in closed position back to the open position.

[0089] The connections, operation, and flow of components in the product gas stream **122** and product gas bypass line **123** is the same as described for apparatus **200** in FIG. 2, and such description is not reproduced here. Operation of the back pressure devices **240** and **244** can be the same as described for apparatus **200**, and such description is not reproduced here.

[0090] The slurry that is removed from the multiphase separator **230** in slurry stream **232** can flow to the sand removal system **250**. The inlet **251** of the sand removal system **250** can be fluidly connected to the slurry stream **232**. The sand removal system **250** can be configured to receive the slurry from the slurry stream **232** and to separate the slurry into the residual gas stream **124**

containing residual gaseous components that were not removed from the slurry in the multiphase separator **230**, a liquid aqueous stream **126**, and a solids stream **128**. In aspects, the residual gas stream **124** contains less than 5, 4, 3, 2, or 1 vol % of the total gases recovered from the second stream **222**. The liquid aqueous stream **126** can contain water, treatment additives (e.g., chemicals injected into the subterranean formation **104** to alter a chemistry within the formation to enhance hydrocarbon recovery), or combinations thereof. Because any condensates and/or crude oil are recovered in liquid hydrocarbon stream **125**, the liquid aqueous stream **126** in apparatus **500** does not contain any significant amount of liquid hydrocarbons (e.g., less than 1, 0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2, or 0.1 wt % liquid hydrocarbons based on a total volume of the liquid aqueous stream **126**). The solids stream **128** can contain sand, and in some aspects, small fragments of rock and plugs that pass through the screen(s) in the plug catcher **210**.

[0091] The sand removal system **250** can be embodied as described for the apparatus **200** in FIG. 2, and such description is not reproduced here, except that since the multiphase separator **230** in apparatus **500** recovers the liquid hydrocarbons from the second stream **222**, there is no need for equipment in the sand removal system **250** or downstream of the sand removal system **250** to recover liquid hydrocarbons from the slurry that is received by the sand removal system **250**, since the slurry contains no significant amount of liquid hydrocarbons.

[0092] FIG. 6 illustrates a cut-away schematic diagram of the embodiment of the multiphase separator **230** in the apparatus **500** of FIG. 5. Parts of the multiphase separator **230** in FIG. 6 are labeled with the same references numerals for like parts illustrated in FIG. 2, FIG. 3, FIG. 4, and FIG. 5. Angled plates **310**, **320**, and **330** can be seen in the interior of the vessel **231** of the multiphase separator **230**.

[0093] The multiphase separator **230** in FIG. 3 is described with respect to separation of the second stream **222** into a first portion of light hydrocarbons in gas phase and a second portion containing liquid hydrocarbons, water, and sand. The second portion can also include treatment additive(s) described herein. The liquid hydrocarbons can include condensates, crude oil, or combinations thereof.

[0094] As previously described for the multiphase separator **230** in FIG. 5, the multiphase separator **230** in FIG. 6 can include a vessel **231**, an inlet **233** connected to a side **236a** of the vessel **231**, a first outlet **234a** connected to a top **237** of the vessel **231**, a second outlet **234b** connected to a bottom **238** of the vessel **231**, and a third outlet **234c** connected to the side **236b** (or side **236a**) of the vessel **231**. The inlet **233** is configured to receive the second stream **222**, the first outlet **234a** is configured to receive the first portion from the interior of the vessel **231** (gases separated from liquids and solids in the vessel **231**), the second outlet **234b** is configured to receive the water, treatment additives, sand, or combinations thereof of the second portion from the interior of the vessel **231**, and the third outlet **234c** is configured to receive the liquid hydrocarbons of the second portion from the interior of the vessel **231**.

[0095] In the multiphase separator **230** of FIG. 6 (and of FIG. 5), a slurry containing water, treatment additive(s), sand, or combinations thereof is removed from the bottom **238** of the vessel **231** of the multiphase separator **230**, and the liquid hydrocarbons are removed from the side **236b** (or side **236a**) of the vessel **231** of the multiphase separator **230**. In aspects, the liquid hydrocarbons are removed separately from the slurry, e.g., the liquid hydrocarbons are removed via liquid hydrocarbon stream **125** and the slurry is removed via slurry stream **232**. In aspects, the slurry is removed intermittently based on a level of the slurry in the multiphase separator **230**, and the liquid hydrocarbons are removed intermittently based on a level of the liquid hydrocarbons in the multiphase separator **230**.

[0096] The multiphase separator **230** in FIG. 6 has the same configuration of angled plates **310**, **320**, and **330** as described for the multiphase separator **230** in FIG. 3; as such, the description is not reproduced here.

[0097] The multiphase separator **230** in FIG. 6 can include two level sensors, a first level sensor

243 and a second level sensor **247**.

[0098] The first level sensor **243** can be seen in FIG. **6** with float **340** connected to a stem **341** that is connected to the body mount **342**. The body mount **342** is connected to the side **236a** of the vessel **231**. While shown connected to side **236a**, the body mount **342** can be connected to any location on the circumference of the vessel **231**. Generally, the first level sensor **243** is connected to the vessel **231** at a location lower than the lowest angled plate (e.g., the third angled plate **330** in FIG. **3**). The float **340** and stem **341** are sized so as not to contact the bottom surface of the lowest plate (e.g., bottom surface **331** of the third plate **330**). The movement of the float **340** of the first level sensor **243** in FIG. **6** is in direction of arrow A in FIG. **3**. The first level sensor **243** is configured such that the float **340** floats on a top surface of the slurry but does not float within or on top of the liquid hydrocarbons in the vessel **231**. For example, for a slurry that contains both solids (e.g., sand) and liquid (e.g., water, treatment additive(s), or both), the sand can fall to the bottom **238** of the vessel **231** and the liquid can position on top of the sand. The float **340** in FIG. **6** is configured with a density less such that a portion **343** of the float **340** is below the top surface **344** of the liquid in the slurry that is in the vessel **231**. For example, the density of the float **340** can be set with weights (e.g., metal weights, or sand in the interior of the float **340**) such that a bottom portion **343** is below the top surface **344** of the liquid in the slurry and a top portion **345** is above the top surface **344** of the liquid in the slurry. The top portion **345** of the float **340** can extend into the liquid hydrocarbons floating on top of the liquid of the slurry while the bottom portion **343** extends into the liquid of the slurry. Configuring the float **340** with a density such that bottom portion **343** is below the top surface **344** of the liquid of the slurry ensures that the float **340** measures the level of the slurry and the operation of level measurement is not affected by the presence of liquid hydrocarbons in the vessel **231**. The float **340** moves up and down as the level of solids and liquids of the slurry raises and lowers due to incoming fluid from the second stream **222** and due to removal of the slurry from the vessel **231** via the slurry stream **232**.

[0099] The second level sensor **247** can be seen in FIG. **6** with float **540** connected to a stem **541** that is connected to the body mount **542**. The body mount **542** is connected to the side **236b** of the vessel **231**. While shown connected to side **236b**, the body mount **542** can be connected to any location on the circumference of the vessel **231**. Generally, the second level sensor **247** is connected to the vessel **231** at a location lower than the lowest angled plate (e.g., the third angled plate **330** in FIG. **6**) and higher than the first level sensor **243**.

[0100] The float **540** is configured with a density less than liquid hydrocarbons that may be present in the vessel **231** (e.g., condensates, crude oil, or both of the well stream **118** that is/are in liquid phase in the multiphase separator **230**) such that the float **540** can float on a top surface **544** of the liquid hydrocarbons inside the vessel **231**. The float **540** moves up and down as the level of liquid hydrocarbons raises and lowers due to incoming fluid from the second stream **222** and due to removal of the slurry from the vessel **231** via the slurry stream **232** and due to removal of the of liquid hydrocarbons from the vessel **231** via liquid hydrocarbon stream **125**.

[0101] The third outlet **234c** is located on the vessel **231** at a location that is lower than the level sensor **247** and at or slightly above the setpoint level for the slurry in the vessel **231** (e.g., at or slightly above the top surface **344** of the liquid in the slurry in the vessel **231**).

[0102] The multiphase separator **230** can be manufactured of suitable metal, flanges, piping, valving, and combinations thereof. Exemplary internal diameter of the vessel **231** of the multiphase separator **230** includes diameters in the range of from about 36 inches to about 60 inches; alternatively, from about 42 inches to about 54 inches; alternatively, about 48 inches. Exemplary heights of the vessel **231** of the multiphase separator **230** includes heights in the range of from about 108 inches to about 132 inches; alternatively, from about 114 inches to about 126 inches; alternatively, about 120 inches. An exemplary location for the level sensor **243** can be a height on the side **236** of the vessel **231** that is about 3 feet above the bottom **238** of the vessel **231**. An exemplary location for the inlet **233** can be a height on the side **236** of the vessel **231** that is about 8

feet above the bottom 238 of the vessel 231, and the angled plates 310/320/330 are placed at locations in the vessel 231 that have a height above the bottom 238 of the vessel 231 that is less than the height of the inlet 233.

[0103] Inlet 233 and outlets 234a and 234c can each have a nominal pipe size in a range of from about 4 inches to about 8 inches; alternatively, about 6 inches. Outlet 234b can have a nominal pipe size that is greater than the inlet 233 and outlets 234a and 234c, for example, in a range of from about 8 inches to about 12 inches; alternatively, about 10 inches. The diameter of outlet 234a is configured to prevent undue friction pressure losses of product gas stream 122 and also to reduce internal flow velocities of the product gas stream 122 where there could be a potential of entraining a small percentage of the liquid phase in the fluid flow into outlet 234a. The larger diameter (nominal pipe size) of the outlet 234b is configured to prevent solids blockage in the outlet 234b.

[0104] In aspects, the operating conditions in the multiphase separator 230 can be the same as those described for the multiphase separator 230 in FIG. 3, and such description is not reproduced here. Moreover, fluid from stream 222 can flow into the vessel 231 via the inlet 233 and components separate such that the first portion moves into the product gas stream 122 and the second portion passes along the angled plates 310/320/330 as previously described. Level sensors 243 and 247 then control the removal of liquid hydrocarbons and slurry from the vessel 231 as described above.

[0105] In some aspects, the techniques described herein relate to a process including: receiving a well stream 118 from a wellbore 102, removing a first plurality of solids having a particle size greater than a particle size of sand from the well stream 118 to form a first stream 212, controlling a flow of the first stream 212 to form a second stream 222 that includes hydrocarbons, water, and sand, separating the second stream 222 in a multiphase separator 230 into a first portion that includes light hydrocarbons in the gas phase and a second portion that includes liquid hydrocarbons, water, and sand, flowing the first portion from the multiphase separator 230 in a product gas stream 122, wherein the product gas stream has a back pressure device 240 located therein, and maintaining, by the back pressure device 240, a pressure in the multiphase separator 230 in a range of from 15.7 psia to about 500 psia. In some aspects of the process, a second pressure of the second stream 222 is in a range of from about 15.7 psia to about 500 psia. In some aspects of the process, the pressure in the multiphase separator 230 is in a range of from about 15.7 psia to about 150 psia; alternatively, about 50 psia.

[0106] In some aspects, the step of maintaining can include sensing a pressure of the product gas stream 122 or the pressure in the multiphase separator 230, and adjusting a flow of the first portion through the back pressure device 240.

[0107] In some aspects, the process can further include flowing the first portion in the product gas stream 122 to one or more compressors 270, and compressing the first portion in the one or more compressors 270 to form a compressed gas stream 272.

[0108] In some aspects, the process can further include flowing the compressed gas stream 272 to a sales pipeline, a vent, a flare, a gas processing unit, a storage tank, or a combination thereof.

[0109] In some aspects, the process can include flowing, without compression, the first portion in the product gas stream 122 to a sales pipeline, a vent, a flare, a gas processing unit, a storage tank, or a combination thereof.

[0110] In some aspects, the process can include removing the second portion from the multiphase separator 230 (e.g., via the slurry stream 232).

[0111] In some aspects, the process can include flowing the second portion in the slurry stream 232 from the multiphase separator 230 to a sand removal system 250, and separating, in the sand removal system 250, the second portion into a residual gas stream 124, a liquid aqueous stream 126, and a solids stream 128.

[0112] In some aspects, the process can include combining the residual gas stream 124 with at least a portion of a product gas stream 122 to form a combined gas stream; and flowing the combined gas stream to a flare 260.

[0113] In some aspects, the process can include flowing the second portion in the slurry stream from the multiphase separator **230** to a tank in the sand removal system **250**.

[0114] In some aspects of the process, removing the second portion is performed intermittently based on a level of the second portion and sand in the multiphase separator **230**.

[0115] In some aspects, the process can further include actuating a valve **242** from a closed position to an open position when the level of the second portion in the multiphase separator **230** exceeds a setpoint level.

[0116] In some aspects, the process can include selectively flowing the first stream **212** directly to the multiphase separator **230** or to the sand removal system **250**, without passing the first stream **212** to the choke manifold **220**.

[0117] In some aspects of the process, the first plurality of solids are removed in a plug catcher **210**.

[0118] In some aspects, the process can include removing a slurry comprising the water and sand from the multiphase separator **230**; and removing, separately from the slurry, the liquid hydrocarbons from the multiphase separator **230**.

[0119] In some aspects of the process, removing the slurry is performed intermittently based on a level of the slurry in the multiphase separator **230**, wherein removing the liquid hydrocarbons is performed intermittently based on a level of the liquid hydrocarbons in the multiphase separator **230**.

[0120] In some aspects, the process can additionally include flowing the slurry in a slurry stream **232** from the multiphase separator **230** to a sand removal system **250**; and flowing the liquid hydrocarbons in a liquid hydrocarbon stream **125** from the multiphase separator **230**.

[0121] In some aspects, the process can additionally include separating, in the sand removal system **250**, the slurry into a residual gas stream **124**, a liquid aqueous stream **126**, and a solids stream **128**.

[0122] In aspects, the process is performed with any embodiments described herein for the multiphase separator **230**. Additionally or alternatively, the process is performed with any embodiments described herein for the plug catcher **210**, the choke manifold **220**, the back pressure device **240**, the control valve **242**, the back pressure device **244**, the sand removal system **250**, the flare **260**, the compressor(s) **270**, or combinations thereof.

[0123] Although the present disclosure and its advantages have been described in detail, it should be understood that various changes, substitutions and alterations can be made herein without departing from the spirit and scope of the disclosure as defined by the appended claims. Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, composition of matter, means, processes and steps described in the specification. As one of ordinary skill in the art will readily appreciate from the disclosure, processes, machines, manufacture, compositions of matter, means, processes, or steps, presently existing or later to be developed that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein may be utilized according to the present disclosure. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, processes, or steps.

Claims

1. A process comprising: receiving, during flowback before production, a well stream from a wellbore; removing, during flowback before production, a first plurality of solids having a particle size greater than a particle size of sand from the well stream to form a first stream; controlling, during flowback before production, a flow of the first stream to form a second stream comprising hydrocarbons, water, and sand; separating, during flowback before production, the second stream in a multiphase separator into only a first portion and a second portion, wherein the first portion

comprises light hydrocarbons in a gas phase, wherein the second portion comprises liquid hydrocarbons, water, and sand; flowing, during flowback before production, the first portion from the multiphase separator in a product gas stream; flowing, during flowback before production, the second portion in a slurry stream from the multiphase separator to a sand removal system; and separating, in the sand removal system during flowback before production, the second portion into a liquid aqueous stream comprising the liquid hydrocarbons and the water of the second portion and solids comprising the sand of the second portion, wherein the sand removal system is downstream of the multiphase separator.

2. The process of claim 1, wherein the first plurality of solids comprise drill cuttings, isolation plug remnants, or combinations thereof.
3. The process of claim 1, wherein the multiphase separator is a vertical separator.
4. The process of claim 1, wherein the product gas stream has a back pressure device located therein, the process further comprising: maintaining, by the back pressure device during flowback before production, a pressure in the multiphase separator.
5. The process of claim 4, wherein maintaining comprises: sensing, during flowback before production, a pressure of the product gas stream or the pressure in the multiphase separator; and adjusting, during flowback before production, a flow of the first portion through the back pressure device.
6. The process of claim 4, wherein the pressure maintained in the multiphase separator by the back pressure device is in a range of from 15.7 psia to about 500 psia.
7. The process of claim 1, further comprising: flowing, during flowback before production, the product gas stream to a sales pipeline, a generator that powers equipment, or both.
8. The process of claim 7, wherein the product gas stream is compressed before flowing to the sales pipeline, the generator, or both.
9. The process of claim 1, further comprising: recovering, by the sand removal system during flowback before production, a residual gas stream from the second portion; and flowing, during flowback before production, the residual gas stream to a flare.
10. The process of claim 1, wherein the multiphase separator comprises: a vertical vessel; an inlet connected to a side of the vertical vessel; a first outlet connected to a top of the vertical vessel; a second outlet connected to a bottom of the vertical vessel, wherein the inlet is configured to receive the second stream, wherein the first outlet is configured to receive the first portion, and wherein the second outlet is configured to receive the second portion; a plurality of angled plates, wherein each of the plurality of angled plates has an end connected to an inner wall of the vertical vessel; and a level sensor comprising a float, wherein the float is configured to float on a surface of the second portion in the vertical vessel, wherein a top plate of the plurality of angled plates is connected to the inner wall at a location that is below where the inlet is connected to the vertical vessel, wherein an angle between a bottom surface of each of the plurality of angled plates and the inner wall is in a range of from about 45° to about 75°, wherein an opposite end of each of the plurality of angled plates extends past a longitudinal axis of the vertical vessel.
11. The process of claim 1, wherein the light hydrocarbons comprise methane, ethane, propane, butane, or combinations thereof.
12. The process of claim 1, wherein the light hydrocarbons are present in the product gas stream in an amount that is greater than 95 vol % based on a total volume of the product gas stream.
13. The process of claim 1, wherein the liquid hydrocarbons comprise crude oil, condensates, or a combination thereof.
14. The process of claim 1, wherein the sand of the second portion is dispersed in the liquid hydrocarbons and water of the second portion while flowing the second portion in the slurry stream.
15. The process of claim 1, wherein flowing the second portion is performed based on a level of the second portion in the multiphase separator.

- 16.** The process of claim 1, further comprising: controlling, during flowback before production, a flow of the second portion from the multiphase separator based on a level of the second portion in the multiphase separator.
- 17.** The process of claim 16, wherein controlling the flow of the second portion comprises: sensing the level of the second portion in the multiphase separator; sending a control signal to a control valve; and actuating the control valve to an open position or to a closed position to allow or disallow the flow of the second portion based on the control signal, wherein a level sensor is coupled to the multiphase separator, wherein the control valve and the level sensor are operably coupled to a control system that is configured to i) receive a sensor signal from the level sensor, ii) analyze the sensor signal, and iii) send the control signal to the control valve.
- 18.** The process of claim 1, wherein flowback occurs after a treatment of a subterranean formation that is fluidly connected to the wellbore.
- 19.** The process of claim 18, wherein the treatment comprises fracturing the subterranean formation.
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